

EXERCISE 6: TESTING & CI/CD SETUP

Aim

To implement testing strategies and understand CI/CD pipeline for ensuring software quality in the Credit Card Processing System.

Tools Used

- Python
- GitHub
- VS Code / Google Colab

Procedure

62. Test Strategy Document Creation

A test strategy was developed to ensure the quality of the application.

The following types of testing were included:

- Unit Testing
- Integration Testing
- User Interface Testing

63. Unit Test Implementation

Unit tests were implemented to verify individual components of the system.

A basic test case was created to validate payment processing logic.

64. Integration Test Implementation

Integration testing was performed to ensure proper interaction between modules such as payment processing and fraud detection.

65. UI Test Implementation

UI testing was considered to verify user interface functionality and ensure proper user interaction.

66. CI/CD Pipeline Setup

A CI/CD workflow was defined to automate software development processes.

The pipeline includes:

- Code Integration
- Automated Testing
- Deployment

67. Test Report Generation

Test reports were generated to evaluate system performance.

The results indicated that:

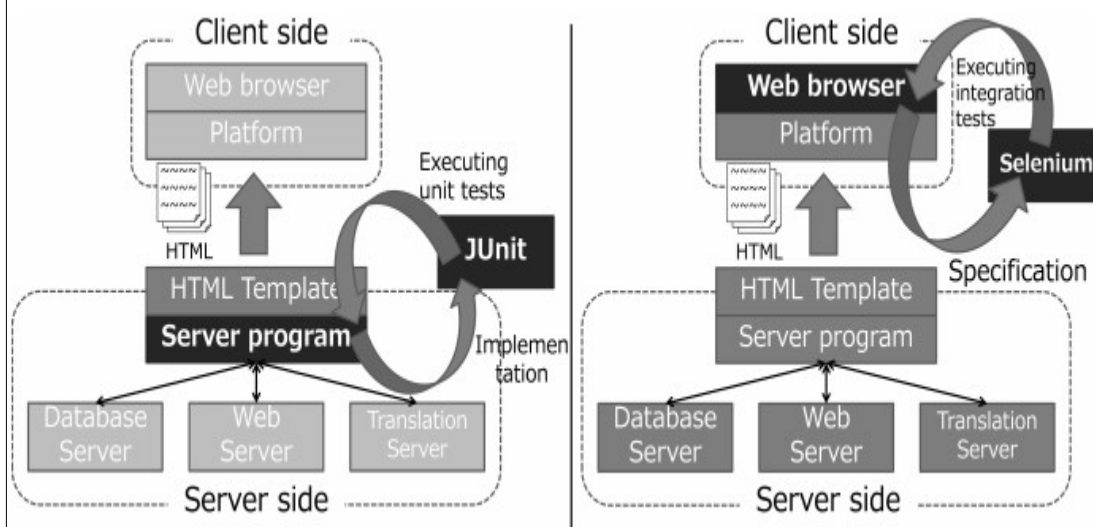
- All test cases passed successfully
- No critical errors were found

68. Defect Metrics Tracking

Defect tracking was performed to monitor system issues.

No major defects were identified during testing.

Figure 6.1: Testing Workflow



Description: This diagram shows the flow of testing and CI/CD process.

Result

Successfully implemented testing strategies and understood CI/CD workflow for ensuring system quality.